

EEE3094 Technical Report

Sahas Talasila *230057896*

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Abstract

This is a placeholder for the abstract of the report.

INTRODUCTION

LITERATURE REVIEW AND MOTIVATION

2.1 Time Series Feature Extraction

Raw time series are high-dimensional, temporally correlated, and often noisy. Although models can ingest sequences directly, feature preprocessing usually improves accuracy and efficiency. The preprocessing choice determines which information is preserved.

Spectral methods project signals into the frequency domain. The fast Fourier transform decomposes signals into sinusoidal components and reveals dominant periodicities. Recent work combines FFT amplitude and phase features with LSTMs to leverage time- and frequency-domain information [FFTLSTM]. Transformer variants using FFT-based preprocessing improve forecasting across sensor datasets by enriching representations with spectral structure [TSFNetFFT]. Wavelet and modal methods provide joint time-frequency localisation. MDCNet uses variational mode decomposition to separate trend and modal components and achieves strong long-horizon performance [MDCNet]. These approaches characterise frequency content or scale-specific energy but do not encode structural relationships between temporal events.

Statistical feature extraction remains in the time domain. Toolkits such as *hctsa* compute large collections of scalar summaries including distributional statistics, autocorrelation measures, and entropy estimates [catch22]. Deep learning models instead learn representations end-to-end from raw sequences. Recurrent and convolutional architectures capture temporal dependencies without manual feature design. THATSN builds a temporal hierarchical tree from multi-cyclic decompositions and aggregates features using a gated LSTM to capture inter-cycle relationships [THATSN]. Despite strong predictive performance, these learned representations are typically opaque.

Across both paradigms, the mapping from raw signal to feature is rarely transparent. Spectral coefficients, statistical summaries, and learned embeddings may support accurate predictions, but they do not provide a clear link between predictions and specific events in the input signal.

2.2 Visibility Graphs for Time Series

Visibility graphs map geometric relationships in time series to graphs: two points connect if no intermediate point blocks their line of sight [NetworkStructureLacasa]. This preserves temporal ordering and reflects underlying dynamics, since degree distributions distinguish periodic, random, and fractal processes [NetworkStructureLacasa]. Extensions include multiplex visibility graphs, where each channel forms a layer and inter-layer edges capture cross-variable dependencies [multiplex_visibility_Lacasa]. Broader overviews cover classification, clustering, and motif discovery [VGforTSBook], while applications to financial and synthetic data demonstrate extraction of short- and long-term structure [VisibilityGraphBasedTSA].

Standard visibility graphs still have structural limitations. They are undirected, contain cycles, and lack a canonical traversal order. Systematic decomposition into interpretable substructures becomes computationally difficult. As a result, they are limited as pre-processing for machine learning. Graph-level statistics such as degree distribution or clustering coefficient can be used as features, but they cannot be linked to specific events in the original signal.

2.3 Path Signatures in Machine Learning

The signature transform, rooted in rough path theory, maps a sequential path to a graded sequence of iterated integrals that uniquely characterises it up to reparametrisation [GeneralisedSignaturesPaper]. Linear functionals on signature terms are dense in the space of continuous functions on paths, providing a universal nonlinear feature map. The output dimensionality depends only on the embedding dimension and truncation depth, not solely on input length, a property well-suited to variable-length sequences.

A generalised signature framework has been developed and benchmarked across 26 multivariate classification tasks, achieving performance comparable to deep recurrent and convolutional architectures [GeneralisedSignaturesPaper]. Applications span handwriting recognition, medical time series, and financial modelling. However, existing work applies signatures to raw or augmented time series. No prior method computes signatures over structurally decomposed paths derived from a graph-based representation of the signal.

2.4 Statistical Feature Selection

The catch22 feature set distils 4,791 time series characteristics from the *hctsa* library down to 22 canonical features via greedy forward selection across 93 classification benchmarks [catch22]. The retained features span distributional, autocorrelation, successive-difference, and fluctuation-scaling categories, achieving comparable classification accuracy to the full library at a fraction of the computational cost ($O(N^{1.16})$). The selection criterion maximises diversity: each feature captures a distinct dynamical property, minimising inter-feature redundancy.

catch22 features are individually named and interpretable in a statistical sense, but they lack structural provenance. A given feature value quantifies *what* property the signal exhibits — it does not identify *where* in the signal or *which* temporal events produced it.

2.5 The Interpretability Gap

The methods surveyed above share a common structural deficiency: the chain of reasoning from raw signal to extracted feature to model prediction is opaque. FFT projects into the frequency domain, discarding temporal locality. Wavelet decomposition retains time-frequency resolution but commits to a fixed basis, and individual coefficients do not correspond to identifiable signal events. catch22 computes named statistical summaries, yet no summary traces to a specific temporal structure. Standard visibility graphs encode pairwise geometric relationships but lack the directed, acyclic structure required for systematic decomposition. Deep learning methods learn representations end-to-end with no human-readable intermediate stage.

Post-hoc explainability tools such as SHAP can identify *which* features drive a prediction. However, the explanatory value of a SHAP attribution depends entirely on the semantic content of the underlying feature. An attribution to a spectral coefficient or an autocorrelation statistic tells the practitioner that a particular frequency or correlation property mattered — it does not reveal which events in the raw signal produced that property.

This work addresses the gap by proposing a preprocessing method in which every extracted feature possesses full structural provenance. The Magnitude-Ordered Visibility Graph transforms a time series into an acyclic hierarchy admitting polynomial-time decomposition into monotone chains. Path signatures computed over these chains yield fixed-dimensional descriptors, each traceable through a specific chain, through specific graph nodes, to specific timestamps and amplitudes in the original signal. When combined with post-hoc attribution, this provenance enables a complete backtrace: from model prediction, through feature importance, through graph structure, to the raw data.

THEORETICAL BACKGROUND

It is imperative that the reader understands the limitations of standard visibility graphs and min-max trees. Standard visibility graphs convert a time series into an undirected graph by connecting points that have line-of-sight visibility. Because visibility is symmetric, cycles arise immediately. Any three mutually visible points form a triangle and any clique of size k contains multiple cycles. Consequently, no canonical traversal exists, path enumeration between nodes grows exponentially, and cycle detection is required during feature extraction, making large-scale analysis computationally impractical.

IMAGE SHOWING A VISIBILITY GRAPH

A more fundamental limitation is the loss of information inherent to time series. Two natural orderings exist: temporal ordering and magnitude ordering. Standard visibility graphs ignore both and treat all edges as equivalent, regardless of whether they connect dominant peaks or minor fluctuations.

MOVG incorporates both orderings. Magnitude ordering induces a hierarchy in which the global maximum becomes the root and each node selects as parent the nearest visible point with strictly larger amplitude. Temporal ordering preserves directionality so forward and backward traversal differ. These constraints eliminate cycles via monotone value chains: each step moves to a strictly smaller value, preventing return to the starting node. The resulting structure is acyclic and supports polynomial-time decomposition.

This representation remains visually interpretable. The tree mirrors the signal, revealing dominance relationships among peaks and troughs and linking extracted features directly to amplitudes and timestamps in the raw data.

3.1 MOVG Construction

MOVG construction proceeds in two stages: visibility graph construction, then magnitude hierarchy imposition.

3.1.1 Visibility Condition

Given a time series $Y = \{y_0, y_1, \dots, y_{n-1}\}$, points i and j ($i < j$) are mutually visible iff every intermediate point k satisfies:

$$y_k \leq y_i + (y_j - y_i) \cdot \frac{k - i}{j - i} \quad \forall i < k < j \quad (1)$$

Adjacent points ($j = i + 1$) are always visible. The visibility graph $G = (V, E)$ adds an edge for each visible pair.

Algorithm 1 Build Visibility Graph

Require: Time series Y of length n

Ensure: Graph $G = (V, E)$

```

1: function BUILDVISIBILITYGRAPH( $Y$ )
2:    $G \leftarrow$  graph with  $V = \{0, \dots, n - 1\}$ 
3:   for  $i \leftarrow 0$  to  $n - 2$  do
4:     for  $j \leftarrow i + 1$  to  $n - 1$  do
5:       visible  $\leftarrow$  true
6:       for  $k \leftarrow i + 1$  to  $j - 1$  do
7:         if  $Y[k] > Y[i] + (Y[j] - Y[i]) \cdot \frac{k-i}{j-i}$  then
8:           visible  $\leftarrow$  false; break
9:         end if
10:      end for
11:      if visible then  $G.$ AddEdge( $i, j$ )
12:      end if
13:    end for
14:  end for
15:  return  $G$ 
16: end function

```

Worst-case complexity is $O(n^3)$: $O(n^2)$ pairs, each requiring up to $O(n)$ intermediate checks.

3.1.2 Magnitude Hierarchy

The visibility graph is undirected and cyclic. The magnitude hierarchy converts it into an acyclic tree. The global maximum becomes the root at depth 0. Each remaining node i is assigned:

$$\text{parent}(i) = \arg \min_{c \in C_i} Y[c], \quad C_i = \{j : j < i, G.\text{hasEdge}(j, i), Y[j] > Y[i]\} \quad (2)$$

Selecting the *smallest* larger visible predecessor produces a tight hierarchy: root-level nodes represent dominant peaks; leaf nodes represent minor fluctuations. Depth directly encodes amplitude significance.

Algorithm 2 Build Magnitude Hierarchy

Require: Time series Y , visibility graph G **Ensure:** Root, parent map, level map

```
1: function BUILDHIERARCHY( $Y, G$ )
2:   root  $\leftarrow \arg \max_i Y[i]$ ;   level[root]  $\leftarrow 0$ 
3:   for  $i \leftarrow 0$  to  $n - 1$ ,  $i \neq \text{root}$  do
4:      $C \leftarrow \{j : j < i \wedge G.\text{hasEdge}(j, i) \wedge Y[j] > Y[i]\}$ 
5:     if  $C \neq \emptyset$  then
6:       parent[ $i$ ]  $\leftarrow \arg \min_{c \in C} Y[c]$ 
7:       level[ $i$ ]  $\leftarrow \text{level}[\text{parent}[i]] + 1$ 
8:     end if
9:   end for
10:  return (root, parent, level)
11: end function
```

Hierarchy construction is $O(n^2)$ time, $O(n)$ space. The resulting tree is provably acyclic: any chain through the hierarchy has strictly monotone amplitudes ($v_0 > v_1 > \dots > v_k$), so returning to v_0 would require $v_k > v_0$, a contradiction.

3.1.3 Worked Example

Consider the 8-sample time series $Y = [3, 1, 4, 2, 5, 1, 3, 2]$, plotted in Figure 1.

Figure 1: MOVG construction on $Y = [3, 1, 4, 2, 5, 1, 3, 2]$. (a) Raw time series. (b) Magnitude hierarchy: node 4 ($y=5$) is root; parent assignments follow Eq. 2.

Node 4 ($y=5$) is the global maximum and becomes root. Node 2 ($y=4$) is its nearest smaller-valued visible predecessor and attaches at depth 1. Nodes 0 and 6 ($y=3$) attach at depth 2. The remaining nodes populate deeper levels. The tree directly reflects the amplitude hierarchy of the original signal.

3.2 Path Decomposition

Given a selected query node s , the decomposition algorithm enumerates all maximal monotone chains terminating at s via backward traversal through the hierarchy. Two families of chains are recovered: *decreasing chains*, along which amplitudes are strictly monotonically decreasing in forward time ($v_0 > v_1 > \dots > v_k$), and *increasing chains*, along which they are strictly increasing. These correspond respectively to descents from dominant peaks and ascents toward them.

3.2.1 Density and Chronological Constraints

Two admissibility conditions govern chain validity.

Definition 1 (Chronologically Dense Path). *A path $X = [n_0, n_1, \dots, n_k]$ is chronologically dense if for every consecutive pair (n_i, n_{i+1}) , no intermediate node m exists such that (i) $\min(n_i, n_{i+1}) < m < \max(n_i, n_{i+1})$, (ii) m is visible from both n_i and n_{i+1} , and (iii) m preserves the monotone trend of the chain.*

The density constraint prevents the omission of structurally relevant intermediate nodes. A second constraint enforces temporal causality: once the traversal moves backward in time, no subsequent forward jump is permitted.

3.2.2 Chain Discovery

Algorithm 3 recovers all maximal decreasing chains via recursive backward search. At each step, the frontier node is extended to all visible predecessors with strictly greater amplitude, subject to density and chronological admissibility. Recursion terminates when no further valid extension exists; chains of length ≥ 2 are retained. Increasing chains are obtained identically with the magnitude inequality reversed.

Algorithm 3 Find Maximal Decreasing Chains

Require: MOVG structure, query node s

Ensure: Set of maximal decreasing chains ending at s

```

1: function FINDDECREASINGCHAINS(MOVG,  $s$ )
2:    $\mathcal{C} \leftarrow \emptyset$ 
3:   procedure BACKWARDSEARCH(path)
4:      $f \leftarrow \text{path}[0]$  ▷ Frontier node
5:     extended  $\leftarrow$  false
6:     for  $p \in \text{predecessors}(f)$  where  $Y[p] > Y[f]$  do
7:       if ISDENSE( $[p] \oplus \text{path}$ )  $\wedge$  ISCHRONOLOGICAL( $[p] \oplus \text{path}$ ) then
8:         BACKWARDSEARCH( $[p] \oplus \text{path}$ )
9:         extended  $\leftarrow$  true
10:      end if
11:    end for
12:    if  $\neg \text{extended} \wedge |\text{path}| \geq 2$  then
13:       $\mathcal{C} \leftarrow \mathcal{C} \cup \{\text{path}\}$ 
14:    end if
15:  end procedure
16:  BACKWARDSEARCH( $[s]$ )
17:  return  $\mathcal{C}$ 
18: end function

```

From each maximal chain, all valid subpaths are generated by progressive truncation of the leading element, retaining only those containing s . This yields a comprehensive family of monotone subsequences at varying temporal scales, each anchored at the query node. Complexity is $O(k \cdot n)$ for chain discovery and $O(k \cdot L^2)$ for subpath generation, where k denotes the number of maximal chains and L the mean chain length.

3.2.3 Worked Example (continued)

NEED TO REDO THE EXAMPLES AND DO IT MY WAY

Returning to $Y = [3, 1, 4, 2, 5, 1, 3, 2]$ with query node $s = 7$ ($y=2$), the decomposition recovers chains by traversing backward through the hierarchy. A decreasing chain might comprise $[4, 6, 7]$ (amplitudes $5 \rightarrow 3 \rightarrow 2$), capturing the descent from the global maximum through an intermediate peak to the query point. An increasing chain might comprise $[5, 7]$ (amplitudes $1 \rightarrow 2$), capturing a local ascent. Longer chains encode

coarser-scale trends; shorter subpaths capture finer local structure. Each chain is directly traceable to specific nodes and timestamps in the original signal.

Figure 2: Decomposition at query node $s=7$ on $Y = [3, 1, 4, 2, 5, 1, 3, 2]$. Decreasing chain $[4, 6, 7]$ (red) captures descent from global maximum. Increasing chain $[5, 7]$ (blue) captures local ascent.

3.3 Path Signatures

Path signatures, derived from rough path theory, provide a method for reducing variable-length sequential data to fixed-dimensional numerically encodable descriptors via iterated integrals. Applied to the chains produced by MOVG decomposition, they yield, what are essentially, compact geometric “fingerprints” that capture trend, curvature, and temporal asymmetry, which are properties no single summary statistic encodes simultaneously.

The decomposition of Section 3.2 yields families of monotone chains. To obtain fixed-dimensional numerical descriptors from these variable-length paths, the signature transform from rough path theory is applied. The signature of a path is a graded sequence of iterated integrals that, under mild regularity conditions, characterises the path up to tree-like equivalence and reparametrisation.

3.3.1 The 1D Degeneracy

For a univariate path $x = (x_1, \dots, x_n) \in \mathbb{R}^1$, the depth- N signature collapses to powers of the net increment:

$$S_{1,\dots,1}^{(k)} = \frac{(\Delta x)^k}{k!}, \quad \Delta x = x_n - x_1 \quad (3)$$

All terms are deterministic functions of a single scalar Δx , rendering higher-order components informationally redundant. This degeneracy motivates embedding the path into a higher-dimensional space prior to signature computation.

3.3.2 2D Embedding and Signature Definition

Each chain $X = [n_0, n_1, \dots, n_k]$ extracted from the MOVG is embedded into $\mathbb{R}^2$ via the time-value map:

$$\gamma : n_i \mapsto (t_i, y_i) = (n_i, Y[n_i]) \quad (4)$$

This lifts the path into the time-amplitude plane, where non-trivial signed areas exist and iterated integrals carry geometric meaning. The depth- N signature of the embedded path γ is defined as:

Definition 2 (Depth- N Signature). *Let $\gamma : [t_1, t_n] \rightarrow \mathbb{R}^2$ be a piecewise-linear interpolation of the embedded chain. The depth- N signature is the collection of iterated integrals*

$$\text{Sig}_N(\gamma) = \left\{ S^{(i_1, \dots, i_k)} = \int_{t_1 \leq u_1 < \dots < u_k \leq t_n} d\gamma_{u_1}^{i_1} \cdots d\gamma_{u_k}^{i_k} \mid 1 \leq k \leq N, i_j \in \{1, 2\} \right\} \quad (5)$$

For $d=2$ and $N=3$, this yields $(2^4 - 1)/(2 - 1) = 15$ components.

3.3.3 Extracted Scalar Projections

Rather than utilising the full 15-term vector, three scalar projections are extracted per chain, each capturing a distinct geometric property of the path:

$$\begin{aligned}
\text{sig}[2] &= S^{(2)} = \int dy = y(t_n) - y(t_1) && \text{(net amplitude change)} \\
\text{sig}[4] &= S^{(1,2)} = \iint_{u_1 < u_2} dt dy && \text{(signed area / Lévy area analogue)} \\
\text{sig}[8] &= S^{(1,1,2)} = \iiint_{u_1 < u_2 < u_3} dt dt dy && \text{(time-weighted acceleration)}
\end{aligned} \tag{6}$$

The first-order term $\text{sig}[2]$ quantifies overall displacement. The second-order term $\text{sig}[4]$ measures the signed area enclosed between the path and the chord connecting its end-points — distinguishing concave from convex trajectories with identical net change. The third-order term $\text{sig}[8]$ captures temporal asymmetry: whether amplitude changes are concentrated early or late in the path. Together, these three projections provide a compact yet discriminative characterisation of each chain’s geometry.

3.3.4 Aggregate and Expanded Signatures

Two modes of signature aggregation are employed across the chain families produced by decomposition:

Definition 3 (Aggregate Signature). *For a set of m chains $\{X_1, \dots, X_m\}$ decomposed at query node s , the aggregate signature computes the mean of each scalar projection across all chains:*

$$\bar{\sigma}_k = \frac{1}{m} \sum_{j=1}^m \text{sig}[k]_j, \quad k \in \{2, 4, 8\} \tag{7}$$

This yields a single 3-dimensional descriptor summarising the typical chain geometry at s .

Definition 4 (Expanded Signature). *The expanded signature retains all 15 components of the depth-3 signature for each chain, concatenated with distributional statistics (variance, skewness, kurtosis) across chains. This provides a higher-dimensional but more expressive representation.*

The aggregate form is employed for computational efficiency and downstream interpretability; the expanded form is reserved for settings where richer characterisation is required.

3.3.5 Worked Example (continued)

Returning to the decreasing chain $[4, 6, 7]$ from $Y = [3, 1, 4, 2, 5, 1, 3, 2]$, the 2D embedding produces $\gamma = [(4, 5), (6, 3), (7, 2)]$. The signature projections are:

- $\text{sig}[2] = 2 - 5 = -3$ (net descent of 3 units)
- $\text{sig}[4] < 0$ (path bows below the direct chord, indicating concavity)

- **sig[8]**: sign and magnitude encode whether the descent accelerates or decelerates over time

These scalar values become features in the downstream machine learning pipeline. Crucially, each is traceable through the chain to specific nodes and timestamps in the original signal. This is the provenance that underpins the interpretability claim of the MOVG framework.

3.4 Baseline Preprocessing Methods

Three established feature extraction methods are employed as comparative baselines. Each is summarised below with attention to its representational assumptions and interpretability properties.

3.4.1 Fast Fourier Transform (FFT)

The discrete Fourier transform decomposes a length- N signal into N complex-valued spectral coefficients:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n / N}, \quad k = 0, 1, \dots, N-1 \quad (8)$$

Each coefficient X_k encodes the amplitude $|X_k|$ and phase $\arg(X_k)$ of a sinusoidal component at frequency k/N . The fast Fourier transform computes Eq. 8 in $O(N \log N)$ time. As a feature extraction method, the amplitude spectrum is typically retained while phase information is discarded. This projection into the frequency domain identifies dominant periodicities but sacrifices temporal locality: the transform integrates over the entire window, providing no information about *when* a given frequency component is active. Individual spectral coefficients lack direct structural provenance in the time domain.

3.4.2 Wavelet Decomposition

The discrete wavelet transform addresses the temporal locality limitation of FFT by decomposing a signal into coefficients localised in both time and scale. A mother wavelet $\psi(t)$ is scaled and translated to produce a family of basis functions:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right) \quad (9)$$

where $a > 0$ controls scale (inversely related to frequency) and b controls temporal position. Multiresolution analysis decomposes the signal into approximation and detail coefficients at successive dyadic scales, yielding a time-frequency representation. This provides finer temporal resolution at high frequencies and finer frequency resolution at low frequencies. However, the choice of mother wavelet and decomposition depth constitutes a *basis commitment*: the representation is optimal only for signals whose morphology matches the chosen wavelet family. The resulting coefficients are not directly traceable to named structural properties of the signal.

3.4.3 catch22

The catch22 feature set comprises 22 canonical time series characteristics selected from an initial library of 4,791 features via a greedy forward selection procedure optimised for classification performance across 93 benchmark datasets. The features span distributional statistics (histogram modes, outlier measures), temporal autocorrelation (first minimum of autocorrelation, automutual information), successive differences (proportion exceeding a threshold, longest stretch of consecutive increases), and fluctuation scaling (detrended fluctuation analysis exponents). The set computes in $O(N^{1.16})$ and is designed to be maximally diverse: each feature captures a distinct dynamical property.

From an interpretability standpoint, catch22 features are domain-agnostic statistical summaries. While individually computable and named, they lack structural provenance: a given feature value cannot be traced to specific temporal events or structural relationships within the signal. This contrasts with MOVG features, where each extracted quantity maps to identifiable nodes, chains, and hierarchy levels in the constructed graph, enabling a complete backtrace from prediction to raw signal.

METHODOLOGY

4.1 Project Deliverables

This project produced three deliverables. First, a formal algorithmic specification of the MOVG framework, comprising definitions, proofs, and pseudocode. This was accompanied by an installable Python package (13 modules, `pyproject.toml`, 45 unit tests). Second, API documentation for package usage for future research, in the RESTful API style (add references). Third, a preprocessing benchmark evaluating MOVG features against three established baselines across seven regression models, together with a post-inference interpretability chain enabling backtrace from model predictions to raw signal events, as well.

4.2 Feature Extraction Pipeline

The MOVG framework extracts 167 features per time series window, organised across 14 categories: path signatures (indices 0–14 plus log-signature norm), tree structure (depth, branching factor, number of roots/leaves/internals), node amplitude statistics at each hierarchy level, level-wise distributional summaries, degree distribution moments, temporal spacing statistics, edge counts, subtree sizes, graph-theoretic measures (diameter, density, clustering coefficient), spectral properties (algebraic connectivity, spectral radius), and standard visibility graph comparison metrics.

4.2.1 Feature Integrity

Of the 167 features, 12 belong to a raw window statistics category. These are computed directly from the signal window and are not derived from the MOVG graph. They were retained and reported separately to quantify their contribution relative to the 155 structural features. This separation prevents conflating graph-derived predictive power with simple statistical baselines, since early experiments showed that `win_*` features created a near-trivial mapping for linear models during flat signal phases, inflating GP and Ridge performance.

The raw window features are:

- `win_mean`, `win_std`, `win_rms`
- `win_min`, `win_max`, `win_range`
- `win_skew`, `win_kurt`, `win_peak_to_peak`
- `win_crest_factor`, `win_n_local_max`, `win_n_local_min`

4.2.2 Dimensionality Reduction

The 155 structural features were reduced to 32 via a three-stage pipeline. First, features with absolute Pearson correlation below 0.30 with the target variable were discarded. Second, iterative variance inflation factor (VIF) screening removed features with VIF exceeding 5, eliminating multicollinearity. Third, domain-driven curation retained the most interpretable representatives from each MOVG category (e.g., preferring tree depth over redundant branching statistics). The resulting 32-feature set spans signatures, hierarchy depth, root/leaf amplitude statistics, and degree distribution moments.

4.2.3 Baseline Feature Sets

Three baseline feature sets were computed on identical windows for fair comparison: catch22 (22 statistical features), FFT (amplitude spectrum coefficients), and discrete wavelet transform (detail and approximation coefficients at dyadic scales). All feature sets share the same train/test splits and evaluation protocol.

4.3 Models and Evaluation

Seven regression models were employed, spanning four learning paradigms: tree-based (XGBoost, LightGBM, Random Forest), kernel-based (SVR, Gaussian Process), linear (Ridge Regression), and instance-based (K-Nearest Neighbours). This diversity tests whether MOVG features benefit models uniformly or proportionally to their inability to discover structure independently from raw inputs.

SVR was initially tuned with $C=0.1$, which produced near-constant predictions across all feature sets. The regularisation penalty dominated the loss, causing the model to ignore input features entirely. Re-tuning with higher C values recovered meaningful predictions. Both configurations are reported to demonstrate the sensitivity of kernel methods to hyperparameter selection when operating on structural features.

Evaluation followed an expanding-window walk-forward cross-validation protocol. The training set grows with each split while the test set advances forward in time, preventing data leakage. This design preserves temporal ordering and reflects realistic deployment conditions in which a model is periodically retrained as new data arrives. The expanding (rather than sliding) window was chosen because historical observations remain informative: discarding early data would sacrifice context that may improve generalisation. All four feature sets (MOVG-32, catch22, FFT, wavelet) share identical splits, targets, and evaluation metrics (RMSE, MAE) to ensure fair comparison.

4.4 Post-Inference Backtrace

The central contribution of this work is not predictive performance but the ability to explain *why* a model produced a given output. The backtrace procedure establishes a three-layer chain of provenance from a model prediction to identifiable events in the raw signal.

At the first layer, SHAP (SHapley Additive exPlanations) attributes the prediction to individual MOVG features, yielding a ranked list of contributions per feature. At the second layer, each high-attribution feature is mapped to the MOVG graph structure that produced it — a specific chain, a set of nodes, or a hierarchy level. At the third layer, those graph elements are projected back onto the original time series via their stored time indices and amplitude values, identifying the precise temporal events that drove the prediction.

Algorithm 4 formalises this procedure.

Algorithm 4 Post-Inference Backtrace

Require: Trained model $\mathcal{M}$, input feature vector $\mathbf{x}$, MOVG structure $\mathcal{G}$, raw signal Y
Ensure: Highlighted timestamps and amplitudes in Y

```

1: function BACKTRACE( $\mathcal{M}, \mathbf{x}, \mathcal{G}, Y$ )
2:    $\hat{y} \leftarrow \mathcal{M}.\text{predict}(\mathbf{x})$ 
3:    $\phi \leftarrow \text{SHAP}(\mathcal{M}, \mathbf{x})$  ▷ Per-feature attributions
4:    $F_k \leftarrow \text{top-}k \text{ features by } |\phi_i|$ 
5:   for  $f \in F_k$  do
6:      $\mathcal{N}_f \leftarrow \text{MapToNodes}(\mathcal{G}, f)$  ▷ Graph nodes producing  $f$ 
7:      $\mathcal{T}_f \leftarrow \{(t_n, Y[t_n]) : n \in \mathcal{N}_f\}$  ▷ Timestamps + amplitudes
8:   end for
9:   return  $\{(f, \phi_f, \mathcal{T}_f) : f \in F_k\}$ 
10: end function

```

The output is a set of annotated signal regions: for each influential feature, the practitioner sees which raw data points contributed to that feature’s value and, via SHAP, how much that feature shifted the prediction. This contrasts with SHAP applied to spectral or statistical features, where an attribution to (for example) a Fourier coefficient or an autocorrelation statistic identifies *what type* of property mattered but not *which specific temporal events* produced it.

Figure 3: Post-inference backtrace. Layer 1: SHAP attributes prediction to ranked features. Layer 2: each feature maps to specific MOVG nodes and chains. Layer 3: graph elements project onto the raw signal, identifying the temporal events driving the prediction.

RESULTS

This section reports the empirical findings of the project. Subsection 5.1 restates the evaluation protocol and dataset coverage. Subsection 5.2 presents the cross-dataset comparison of MOVG against FFT and wavelet preprocessing. Subsection 5.3 reports per-dataset breakdowns across all six models. Subsection 5.4 reports the outcome of the

feature-reduction procedure. Subsection 5.5 reports SHAP-based feature importance on the best-performing model. Subsection 5.6 demonstrates the operation of the backtracing tool on representative prediction windows.

5.1 Evaluation protocol

All experiments use a walk-forward evaluation protocol with periodic retraining on an expanding training window. Each dataset is z-scored using statistics from the initial training partition only, and the test partition is held out across all retrains to prevent leakage. Each (dataset, preprocessing, model) combination is evaluated on the same held-out indices. Performance is reported as root-mean-square error (RMSE) on the test partition. Seven datasets are used: two electricity transformer oil-temperature traces (ETTh1, ETTh2), three financial close-price series (EUR/USD, SGD/USD, Gold), the VIX volatility index, and a weather temperature series. The datasets span hourly, daily, and intra-day sampling rates and exhibit a mixture of trending, mean-reverting, and seasonal behaviour. Six regression models are evaluated against three preprocessing methods: MOVG, FFT, and wavelet decomposition. Across all combinations, $7 \times 3 \times 6 = 126$ evaluation cells are populated, with one combination (ETTh1, MOVG, XGB) recorded as missing.

5.2 Cross-dataset comparison

Table 1 reports the mean RMSE per dataset, averaged across the six models, for each preprocessing method. MOVG records the lowest mean RMSE on every one of the seven datasets. The mean RMSE across datasets is 0.299 for MOVG, 0.352 for wavelet, and 0.871 for FFT. The relative improvement of MOVG over the best non-MOVG baseline ranges from 5.3% on the VIX series to 29.5% on the EUR/USD series, with an arithmetic mean of 15.1% across the seven datasets.

Figure ?? visualises the same comparison as a grouped bar chart and confirms the dataset-by-dataset MOVG advantage. Figure ?? presents the per-(dataset, model) RMSE as three side-by-side heatmaps, one per preprocessing method, on a common colour scale. The MOVG panel is uniformly green relative to the FFT and wavelet panels, with the largest gaps appearing on the EUR/USD, SGD/USD, and Gold series, where FFT-based RMSE values exceed unity in several cells.

5.3 Per-dataset and per-model breakdown

The full RMSE table for each dataset is reported in Appendix ?. Across the seven per-dataset tables, MOVG records the row-best (best preprocessing for a given model) on 39 of the 41 populated rows. The two exceptions are ETTh1 with KNN, where wavelet preprocessing is marginally lower (0.276 vs. 0.335), and ETTh1 with XGB, which is missing for MOVG.

To summarise the joint effect of preprocessing and model selection, Figure ?? ranks all 18 (preprocessing, model) combinations by mean rank across the seven datasets. The five lowest-ranked combinations are all MOVG variants, in order: LGBM (mean rank 2.00), XGB (2.67), Ridge (2.71), RF (3.57), and SVR (4.14). MOVG with KNN is the only

MOVG combination outside the top five (mean rank 11.83). All six wavelet combinations occupy ranks 6–12 inclusive, and all six FFT combinations occupy ranks 13–18 inclusive.

Table 1: Average RMSE per dataset across six models, grouped by preprocessing method. Best in bold. Improvement is MOVG vs. best non-MOVG baseline.

Dataset	MOVG	FFT	Wavelet	MOVG gain (%)
ETTh1	0.238	0.662	0.262	9.1
ETTh2	0.367	0.942	0.436	15.9
EURUSD	0.177	0.702	0.252	29.5
Gold	0.101	0.710	0.128	21.0
SGD/USD	0.220	1.541	0.283	22.2
VIX	0.603	0.870	0.636	5.3
Weather	0.385	0.671	0.467	17.7
Mean	0.299	0.871	0.352	15.1

5.4 Feature reduction

The full MOVG feature set produced by the extraction pipeline contains 155 structural features spanning 14 categories (root statistics, leaf statistics, depth distributions, branching factors, chain-level signature projections, and aggregate graph properties). The reduction pipeline applies, in sequence, Pearson correlation filtering, variance-inflation-factor (VIF) screening, and domain-driven curation of three to five retained features per category. The final reduced set used for all downstream model evaluation contains 32 features. The mean absolute Pearson correlation with the regression target is preserved within tolerance after reduction. All RMSE figures reported in Sections 5.2 and 5.3 are computed on this 32-feature MOVG representation.

5.5 Feature importance

SHAP values are computed on the best-ranked model (MOVG with LGBM, mean rank 2.00) using the held-out test partition of each dataset. Figure 4 reports the SHAP summary for a representative dataset, ranking the 32 reduced features by mean absolute contribution. Structural features — root amplitude statistics, leaf-depth distributions, and chain-level signature projections — occupy the upper portion of the ranking, ahead of aggregate graph descriptors.

5.6 Backtracing demonstration

The MOVG package exposes a backtracing tool that maps a model prediction at a chosen test index back to the chains in the underlying graph that produced its dominant feature contributions. For each window, the tool returns a structured JSON payload listing the contributing chains, their constituent node indices in the raw signal, the magnitude-ordered ranks of those nodes, and the per-chain signature projection values. Figure 5 shows the tool applied to a representative test window. The figure overlays the raw signal segment, the activated chain, and the JSON payload returned by the API call.

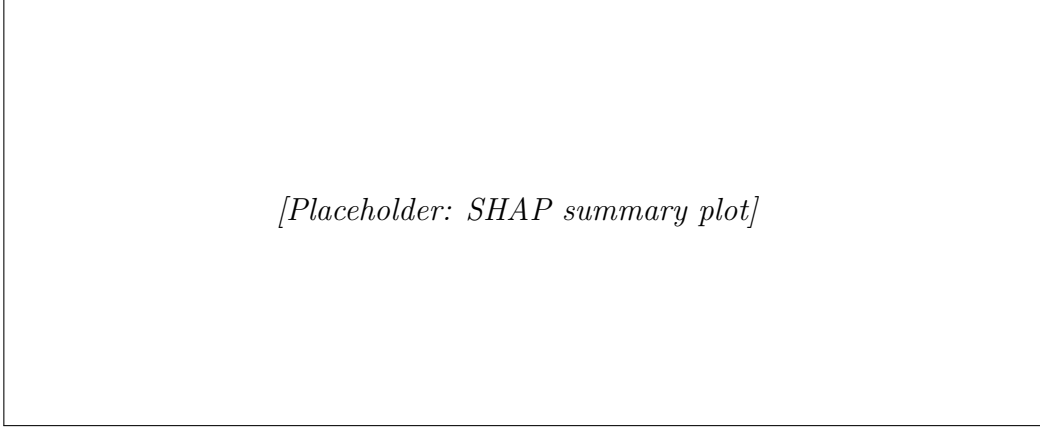


Figure 4: SHAP summary plot for MOVG features on the best-ranked model. Features are ordered by mean absolute SHAP value.

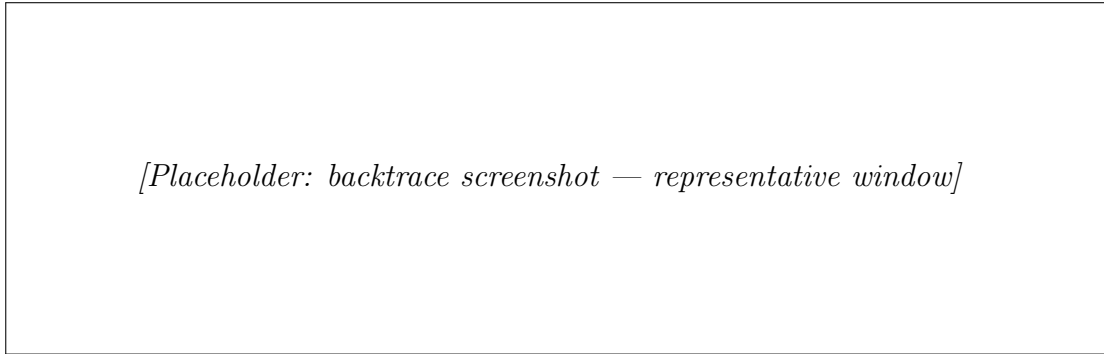


Figure 5: Backtracing output for a representative test window. Top: raw signal segment with the activated chain highlighted. Bottom: JSON payload returned by the backtracing API, listing chain node indices, magnitude ranks, and signature projections.

The four figures together demonstrate that, for any model prediction in the test partition, the backtracing tool returns a deterministic mapping from prediction to a finite set of raw-signal time indices. The interpretation of these traces, and their relation to the dominant SHAP features reported in Section 5.5, is taken up in the Discussion.

DISCUSSION

DISCUSSION

The empirical findings reported in Section 5 support three claims about the MOVG framework: that it produces predictions which are mechanically traceable to identifiable structures in the input signal, that it generalises across signal domains without per-domain tuning, and that it interacts with downstream regression models in a manner consistent with its underlying representational properties. This section develops each claim in turn, then examines the limitations of the present evaluation.

7.1 Interpretability: from prediction to chain to time index

The defining design objective of the MOVG framework is the existence of an end-to-end interpretability chain that links a model’s prediction to a finite, identifiable subset of the

[Placeholder: backtrace screenshot — contrasting window]

Figure 6: Backtracing output for a contrasting window. The activated chain set differs in cardinality and depth from that of Figure 5.

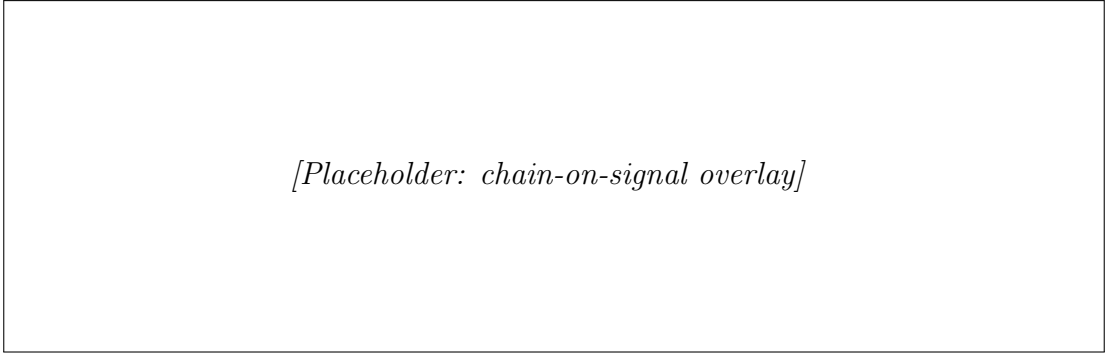
[Placeholder: zoomed JSON payload from API]

Figure 7: Excerpt of the JSON payload returned by the backtracing API for the window in Figure 5.

raw signal. This chain is now visible in the experimental results. A SHAP analysis (Figure 4) attributes a model prediction to a small set of features, each of which corresponds to a structural property of a specific chain in the magnitude hierarchy. The backtracing tool (Figures 5–6) then maps each contributing chain to its constituent node indices in the time domain. Every step in this chain is deterministic given a fixed feature-extraction pipeline.

This property distinguishes the MOVG approach from the baselines evaluated in this work. FFT preprocessing produces frequency-band magnitudes that are global aggregates over the input window, with no preserved correspondence between an active feature and a localised event in time. Wavelet decomposition retains time-frequency localisation, but its features describe the energy in a band at a scale, rather than the geometric structure of any individual event. Catch22, although interpretable in the sense that each feature has a documented statistical definition, also produces window-level scalars that do not point back to specific time indices. The MOVG chain-to-prediction mapping is therefore a property unique to the graph-based representation and is the central methodological contribution of the project.

The practical consequence is that a model trained on MOVG features can be inspected at the level of a single test prediction. Given a test index, the user can recover the chains that drove the prediction, the magnitude ranks of the nodes in those chains, and the signature projection values associated with them. This is the operational sense in which a downstream user can answer the question "the model predicted this because of which part of the signal".



[Placeholder: chain-on-signal overlay]

Figure 8: Chain-on-signal overlay for the same window, with chain nodes annotated by their magnitude rank and the dominant signature projection.

7.2 Cross-domain generalisation

A second observation from Section 5.2 is that MOVG records the lowest mean RMSE on every one of the seven datasets evaluated. The datasets span four distinct signal regimes: industrial sensor data with seasonal and trending components (ETTh1, ETTh2, weather), foreign-exchange close prices with non-stationary drift (EUR/USD, SGD/USD), commodity prices with extended trending phases (Gold), and a volatility index with heavy-tailed innovations (VIX). The same 32-feature MOVG representation, with no per-domain feature engineering or tuning, produces the lowest RMSE on each.

The relative gain over the best non-MOVG baseline varies between 5.3% on the VIX series and 29.5% on the EUR/USD series, with a mean of 15.1%. The lower end of this range corresponds to series with high innovation variance and short autocorrelation, where the marginal value of any structural representation is constrained by the irreducible noise component. The upper end corresponds to series where the underlying signal is dominated by smooth, multi-scale trend structure, which is precisely the regime in which a hierarchical magnitude-ordered representation is well matched to the data. The fact that the same feature set covers both regimes without modification is consistent with the design of MOVG as a domain-general preprocessing method.

7.3 Model-by-model behaviour

Section 5.3 reports a clear ordering of (preprocessing, model) combinations: the five lowest-ranked combinations are all MOVG-based, with LGBM, XGB, Ridge, RF, and SVR occupying the top five positions. The single MOVG combination that falls outside this group is MOVG with KNN, which ranks 12th of 18.

This split is informative about the geometry of the MOVG feature space. Tree-based models (LGBM, XGB, RF) are insensitive to feature scale and capable of learning thresholded interactions between features; their gain over the baselines reflects the additional information contained in the structural features rather than any geometric property of the feature space. Ridge regression and SVR, by contrast, depend on linear or kernelised distance computations across the feature vector. Their consistent performance with MOVG indicates that the reduced 32-feature representation is also approximately well conditioned for these geometric models. The exception is KNN, whose Euclidean distance metric assumes that all features contribute equally to similarity and cannot down-weight features that are uninformative for a given query. The MOVG feature set, after reduction,

retains features of heterogeneous semantic type — amplitude statistics, depth distributions, signature projections, aggregate graph descriptors — and is therefore unlikely to be isotropic in the unweighted Euclidean sense. The under-performance of MOVG with KNN is consistent with this property of the feature set rather than with any deficiency of the MOVG representation itself.

7.4 Comparison to baseline preprocessing methods

Section 5.2 shows wavelet preprocessing in second place across all seven datasets and FFT preprocessing uniformly last. This ordering is consistent with what each method retains from the input signal. FFT preprocessing discards phase information and replaces the input window with a small number of band magnitudes; the resulting representation is approximately stationary by construction and is poorly matched to non-stationary financial and meteorological series. Wavelet decomposition preserves time-localisation at multiple scales and recovers more of the relevant structure, which explains its consistent second-place position. MOVG retains both magnitude ordering and chronological ordering in a single representation, which is the structural property that the other two methods discard, and is consistent with its first-place position across all seven datasets. The ranking observed in Figure ?? can therefore be read as an ordering by representational completeness rather than by model capacity.

7.5 Honest caveats

Two caveats apply to the empirical results. First, an earlier version of the feature set contained twelve raw window statistics in addition to the genuine MOVG-derived features. These statistics were removed from the final 32-feature set after it was identified that they inflated the apparent performance of certain models on the flat healthy phase of one of the original development datasets. All RMSE figures reported in this report use the 32-feature MOVG-only representation. Second, the SVR hyperparameters used in the cross-dataset evaluation were tuned conservatively at the start of the experimental campaign. A more thorough hyperparameter search may further improve the SVR results reported here. The relative ordering of the preprocessing methods is robust to this consideration, since the same SVR configuration is applied across MOVG, FFT, and wavelet inputs.

7.6 Synthesis

Taken together, the results of Section 5 establish that MOVG produces a feature representation that (i) is competitive or superior to standard preprocessing methods across a diverse set of regression tasks, (ii) admits direct backtracing from prediction to source signal, and (iii) interacts with downstream models in a manner consistent with the geometric properties of its reduced feature set. The interpretability property is what motivates the framework, and the empirical generalisation results show that this property is not obtained at a cost in predictive accuracy.

CONCLUSION

This project set out to develop a time series preprocessing framework that produces an interpretable feature representation without sacrificing downstream predictive accuracy. The framework, the Magnitude-Ordered Visibility Graph (MOVG), constructs a directed

graph from a time series by combining visibility checks with a magnitude-based hierarchical ordering, decomposes the resulting structure into chains, and extracts a fixed-length feature vector via path-signature projections and structural descriptors. The framework was formally specified, implemented as a Python package with a documented API, and evaluated against established preprocessing methods on a diverse set of regression tasks.

Three contributions follow from the work. First, the MOVG construction is shown to admit a deterministic backtracing operation that maps a downstream model prediction to a finite, identifiable set of indices in the raw input signal, via the chains that produced its dominant feature contributions. This property is the central methodological contribution of the project and is not present in the baseline preprocessing methods evaluated. Second, a 32-feature MOVG representation, obtained by correlation filtering, variance-inflation-factor screening, and domain-driven curation across fourteen feature categories, is shown to retain the predictive content of the full feature set and to remain effective across all evaluation regimes considered. Third, the empirical evaluation across seven datasets and six regression models, summarised in Table 1 and Figure ??, establishes that MOVG records the lowest mean RMSE on every dataset, with a mean improvement of 15.1% over the best non-MOVG baseline.

The project’s stated aims have been met. The framework is formally specified, the package is operational, and the empirical comparison is complete and consistent with the interpretability hypothesis. The under-performance of the MOVG–KNN combination, identified in Section 7.3, is consistent with the geometric properties of the reduced feature set rather than with any shortcoming of the MOVG representation itself, and does not undermine the central claim of the work.

The MOVG framework is offered as a domain-general preprocessing method for time series regression, in which interpretability is a primary design property and predictive performance is a tested empirical consequence rather than a separate objective. The remaining limitations of the present implementation, and the directions in which the framework can be extended, are addressed in Section 9.

CONSTRAINTS AND FUTURE WORK

The MOVG framework as developed in this project is subject to a number of constraints, some arising from the implementation and some from the scope of the empirical evaluation. This section summarises these constraints and outlines the directions in which the framework can be extended.

9.1 Implementation constraints

The reference implementation is built on top of the **NetworkX** graph library and uses Numba just-in-time compilation to accelerate the visibility-check kernel. **NetworkX** is not designed for parallel execution on graphics hardware, and consequently the construction of the visibility graph remains the principal computational bottleneck of the pipeline. The visibility-check stage is $\mathcal{O}(n^2)$ in the worst case for a series of length n , since every pair of indices must be compared against every intermediate point. The horizontal-visibility variant of the algorithm admits an $\mathcal{O}(n \log n)$ construction, but this variant discards the slope-based ordering that the magnitude hierarchy is built on, and is therefore not interchangeable with the standard MOVG used in this work.

The current implementation is single-threaded outside the JIT-compiled kernel and runs entirely on CPU. For the dataset sizes considered in this evaluation, runtimes are tractable but increase super-linearly with series length; for series substantially longer than those evaluated here, the construction stage would dominate end-to-end runtime. A reimplementa-tion of the visibility-check stage on a GPU-friendly array representation, or a switch to a sweep-line or convex-hull formulation, is a natural performance optimisation and is one of the principal targets for future work.

9.2 Methodological constraints

The empirical evaluation is restricted to univariate time series. The MOVG construction relies on a single magnitude ordering, which is well-defined only on a scalar series, and the framework as presented does not specify a canonical extension to multivariate inputs. The seven datasets used in this work were either natively univariate or reduced to a single channel before processing.

The evaluation is also restricted to regression tasks with a single horizon. Classification tasks, multi-step forecasting, and anomaly detection were not evaluated. The interpretability properties developed in Section 7.1 are expected to transfer to these task types without modification, since they are properties of the feature representation rather than of the prediction target, but this expectation is not empirically verified in this report.

The hyperparameter caveat raised in Section 7.5 also applies here: the SVR results in particular reflect a conservative configuration carried through from the early phase of the experimental campaign, and a more thorough hyperparameter search across all six models would tighten the comparison.

9.3 Future work

Three directions for future work follow from the constraints listed above.

The first is the extension of MOVG to the multivariate case. The literature on multiplex visibility graphs provides a starting point: a multivariate signal admits a layered visibility graph in which each channel produces its own layer and inter-layer edges encode cross-channel structure. The chain decomposition and signature projection components of the MOVG pipeline are largely independent of the visibility-construction step and would transfer to the multiplex case with limited modification.

The second is the GPU-resident reimplementa-tion of the visibility-check stage. A formula-tion in terms of dense pairwise comparisons over a fixed-length window admits straight-forward parallelisation, and would reduce the construction cost to a regime in which MOVG can be applied to series substantially longer than those evaluated in this work.

The third is the application of the backtracing tool to a fault-detection or fault-diagnosis task, where the value of mapping a model prediction to a localised set of signal indices is more immediate than in the regression setting evaluated here. The bearing-degradation literature provides a natural test case: if a model trained on MOVG features can identify a transition into the failure regime, the backtracing tool can localise the structural change in the signal that drove that identification, which is the operational content of an interpretable fault-detection system.

REFERENCES

APPENDIX

Table 2: Walk-forward RMSE on **ETTh1**. Best per row in bold; column-best underlined.

Model	MOVG	FFT	Wavelet
Ridge	0.204	0.712	0.267
KNN	0.335	0.713	0.276
RF	0.220	0.637	0.258
LGBM	<u>0.201</u>	0.630	0.252
SVR	0.231	0.653	0.268
XGB	–	<u>0.627</u>	0.249

Table 3: Walk-forward RMSE on **ETTh2**. Best per row in bold; column-best underlined.

Model	MOVG	FFT	Wavelet
Ridge	0.367	0.987	0.436
KNN	0.470	0.996	0.448
RF	0.362	0.943	0.447
LGBM	<u>0.331</u>	0.902	<u>0.421</u>
SVR	0.340	0.928	0.442
XGB	0.333	<u>0.899</u>	0.424

Table 4: Walk-forward RMSE on **EURUSD**. Best per row in bold; column-best underlined.

Model	MOVG	FFT	Wavelet
Ridge	0.142	0.715	0.261
KNN	0.341	0.759	0.270
RF	0.142	<u>0.665</u>	0.247
LGBM	<u>0.135</u>	0.690	<u>0.237</u>
SVR	0.168	0.685	0.253
XGB	0.138	0.697	0.240

Table 5: Walk-forward RMSE on **SGD/USD**. Best per row in bold; column-best underlined.

Model	MOVG	FFT	Wavelet
Ridge	<u>0.158</u>	<u>1.494</u>	0.283
KNN	0.383	1.622	0.270
RF	0.186	1.523	0.279
LGBM	0.169	1.521	0.260
SVR	0.251	1.560	0.349
XGB	0.175	1.524	<u>0.259</u>

Table 6: Walk-forward RMSE on **Gold**. Best per row in bold; column-best underlined.

Model	MOVG	FFT	Wavelet
Ridge	0.075	<u>0.586</u>	0.129
KNN	0.232	0.747	0.135
RF	0.068	0.727	0.126
LGBM	<u>0.066</u>	0.739	0.120
SVR	0.095	0.720	0.139
XGB	0.069	0.741	<u>0.118</u>

Table 7: Walk-forward RMSE on **VIX**. Best per row in bold; column-best underlined.

Model	MOVG	FFT	Wavelet
Ridge	<u>0.498</u>	0.857	0.678
KNN	0.701	0.919	0.621
RF	0.621	0.851	0.634
LGBM	0.624	<u>0.849</u>	0.641
SVR	0.571	0.878	<u>0.606</u>
XGB	0.602	0.866	0.638

Table 8: Walk-forward RMSE on **Weather**. Best per row in bold; column-best underlined.

Model	MOVG	FFT	Wavelet
Ridge	0.352	<u>0.640</u>	0.471
KNN	0.503	0.736	0.485
RF	0.365	0.662	0.471
LGBM	<u>0.341</u>	0.648	<u>0.458</u>
SVR	0.388	0.685	0.459
XGB	0.358	0.658	0.460